

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems. (L4, L5)

Assessment Tool: Viva Voce

Weightage: 10%

QUESTIONS:

Total Marks: 20

1. Define SOAP.
2. Define WSDL.
3. What is an API?
4. Define XML Schema.
5. What is a Web Service?
6. Define REST.
7. What is a Protocol?
8. Define Endpoint in web services.
9. What is a Service in web applications?
10. What is the difference between SOAP and REST?
11. What is the role of WSDL in web services?
12. What is JSON and how is it used in APIs?
13. What is HTTP and why is it used in web services?
14. What is meant by request and response in web communication?
15. What is the importance of XML in web services?
16. What is the purpose of an API endpoint?
17. What is the difference between HTTP and HTTPS?
18. What is meant by RESTful services?
19. What is a namespace in XML?
20. What is the role of a client and server in web services?

ANSWERS:

1. Define SOAP.

SOAP (Simple Object Access Protocol) is a protocol used for exchanging structured information between applications over a network. It uses **XML-based messages** and supports features such as security, reliability, and error handling.

2. Define WSDL.

WSDL (Web Services Description Language) is an XML-based language used to **describe a web service**. It specifies the operations, messages, data types, communication protocols, and service endpoint.

3. What is an API?

API (Application Programming Interface) is a set of rules and interfaces that allows one software application to communicate with another application or service.

Example: A weather application can use a weather API to obtain current weather data.

4. Define XML Schema.

XML Schema (XSD) is used to define the **structure, elements, attributes, and data types** allowed in an XML document. It is used to validate whether XML data follows the required format.

5. What is a Web Service?

A **Web Service** is a software component that allows different applications to communicate and exchange data over a network using standard web technologies such as **HTTP, XML, JSON, SOAP, or REST**.

6. Define REST.

REST (Representational State Transfer) is an architectural style for designing web services. REST commonly uses HTTP methods such as **GET, POST, PUT, PATCH, and DELETE** to work with resources.

7. What is a Protocol?

A **protocol** is a set of rules that defines how data is **formatted, transmitted, received, and interpreted** between communicating devices or applications.

Examples: HTTP, HTTPS, TCP, FTP.

8. Define Endpoint in Web Services.

An **endpoint** is the specific **URL or network address** through which a web service or API can be accessed.

Example:

<https://example.com/api/patients>

9. What is a Service in Web Applications?

A **service** is a software component that performs a specific function and provides that functionality to other applications or components through a defined interface.

Example: A payment service can process online payments for an application.

10. What is the difference between SOAP and REST?

SOAP	REST
Protocol	Architectural style
Mainly uses XML	Commonly uses JSON
More structured and formal	Lightweight and flexible
Uses SOAP-specific messaging	Uses standard HTTP methods
Supports WS-* standards such as WS-Security	Commonly uses HTTPS, OAuth, JWT, etc.
Generally has more message overhead	Generally lightweight
Suitable for enterprise scenarios requiring strict contracts and specialized WS-* features	Widely used for mobile, web, and cloud APIs

11. What is the role of WSDL in Web Services?

WSDL acts as a **formal contract** between a web service provider and consumer. It describes:

- Available operations
- Input and output messages
- Data types
- Communication binding
- Service endpoint

It helps clients understand **how to communicate with the service**.

12. What is JSON and how is it used in APIs?

JSON (JavaScript Object Notation) is a lightweight text-based data format used to exchange structured data.

Example:

```
{
  "name": "Ravi",
  "age": 22
}
```

APIs commonly use JSON to send **request and response data** because it is compact and easy for applications to parse.

13. What is HTTP and why is it used in Web Services?

HTTP (Hypertext Transfer Protocol) is an application-layer protocol used to transfer data between clients and servers.

It is widely used in web services because it provides standard methods such as:

GET – retrieve data
POST – create/submit data
PUT – replace/update data
PATCH – partially update data
DELETE – remove data

14. What is meant by Request and Response in Web Communication?

Request: A message sent by the client to the server asking it to perform an operation or provide information.

Response: A message returned by the server containing the result of processing the request.

Example:

```
Client
|
| Request: GET /users/101
v
Server
|
| Response: User information
v
Client
```

15. What is the importance of XML in Web Services?

XML (Extensible Markup Language) is used to represent and exchange structured data.

It is important because it:

- Is platform-independent
- Is human-readable
- Supports hierarchical data
- Allows custom tags
- Can be validated using XML Schema
- Is used by SOAP and many related web-service technologies

16. What is the purpose of an API Endpoint?

An **API endpoint** provides a specific location through which a client accesses a particular API resource or operation.

Example:

GET /api/products

This endpoint can be used to retrieve product information.

Different endpoints can represent different resources:

- /api/users
- /api/products
- /api/orders

17. What is the difference between HTTP and HTTPS?

HTTP

Hypertext Transfer Protocol
Data is not encrypted by TLS
Less secure for sensitive information
Uses http://
Does not provide TLS-based server authentication

HTTPS

HTTP Secure
HTTP traffic is protected by TLS
Suitable for sensitive communication
Uses https://
TLS provides server authentication through certificates

HTTPS should be used for sensitive applications such as banking, payments, and login systems.

18. What is meant by RESTful Services?

RESTful services are web services designed according to REST principles.

They generally:

- Use resources identified by URLs
- Use HTTP methods
- Are stateless
- Commonly exchange JSON
- Use standard HTTP status codes

Example:

GET /students → Get students
POST /students → Add student
GET /students/10 → Get student 10
PUT /students/10 → Update student 10
DELETE /students/10 → Delete student 10

19. What is a Namespace in XML?

An **XML namespace** uniquely identifies element and attribute names and prevents naming conflicts between different XML vocabularies.

Example:

```
<soap:Envelope  
  xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
```

Here:

soap is the **prefix**
http://schemas.xmlsoap.org/soap/envelope/ is the **namespace URI**

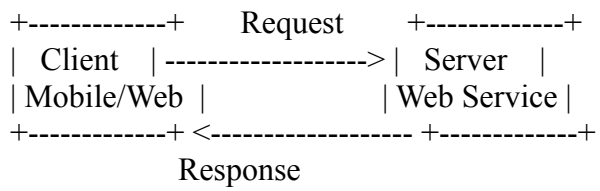
The namespace URI identifies the vocabulary; the prefix is simply a convenient alias.

20. What is the Role of a Client and Server in Web Services?

Client: The application that sends a request to access a service.

Server: The system that receives the request, processes it, and sends a response.

Example:

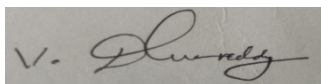


Example: In a banking application, the mobile app is the **client**, while the banking backend/API server is the **server**.

Code of Conduct:

*I V.Dharanireddy(192311219)____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

**RUBRICS FOR CALCULATION:**

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively